

Xtrim Studios — Technical appendix

Methodology, data quality, definitions and the SQL answers behind the Director memo. Generated 14 September 2026.

1. How the work is built

Clean. pipeline/run_pipeline.py reads the 14 raw files (from raw_original_backup/), repairs encodings, parses every date and money format, maps free text to canonical values, converts USD/GBP to naira at the daily rate, deduplicates, and writes clean/ plus a quarantine file. It refuses to write unless rows in = rows out + rows quarantined for every file and every currency column reads NGN. validate.py grades the output against docs/VALIDATION_KEY.md: 93 of 93 checks pass.

Model. warehouse/build_warehouse.py loads clean/ into PostgreSQL 17 (schemas stg, core, quarantine) as a star schema. Primary keys, foreign keys, CHECK constraints and an EXCLUDE constraint on product price windows enforce the grain; a bad load fails.

Answer. warehouse/run_queries.py runs sql/02_questions_answered.sql (14 questions) and saves results/.

Analyse. analysis/analysis.py computes every headline number into analysis/findings.json; charts.py draws only from analysis/tables/. The memo, deck and dashboard read the same files.

Prove it ties. tests/test_tie_out.py checks the dashboard's totals against the SQL results (8 of 8 pass).

Reproduce. One command rebuilds everything: python run_pipeline.py (see README.md).

2. Data quality

282,171 raw rows; 2,445 (0.87%) quarantined with a reason, none deleted. ₦1.44B (6.9%) of revenue is affected by issues that could not be resolved. Full per-column profile: data_quality_report/.

src_table	reason	rows_rejected	value_carried_ngn	revenue_value_at_risk_ngn
bookings	exact_duplicate_booking	30	295,433,323	0.00
bookings	excel_blank Spacer_row	13	0.00	0.00
bookings	duplicate_booking_conflicting_vat_flag	7	162,752,464	0.00
bookings	excel_monthly_subtotal_row	6	0.00	0.00
bookings	excel_footer_note_row	1	0.00	0.00
bookings	excel_grand_total_row	1	0.00	0.00
clients	fuzzy_duplicate_client	124	0.00	0.00
clients	duplicate_client_id_row	58	0.00	0.00
crew_timesheets	orphan_booking_id	319	45,056,500	0.00
crew_timesheets	exact_duplicate_row	223	35,130,500	0.00
marketing_spend	exact_duplicate_row_double_submitted	117	23,418,891	0.00
merch_orders	exact_duplicate_row	786	1,239,281,200	0.00
merch_orders	orphan_client_id	313	179,719,000	179,719,000
merch_orders	placeholder_quantity_9999	190	5,024,300	5,024,300
merch_refunds	orphan_order_id	174	7,600,912	7,600,912

merch_refunds	order_quarantined	83	4,194,450	4,194,450
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Dictionary errors. Fifteen statements in the data dictionary are false, including: client_id is not unique; 2024 booking dates are month-first; the 2025 workbook header is on row 5; timesheet booking ids are not always valid; hours are not capped at 12; the 1.5x rate is unrelated to overtime; price windows overlap; order timestamps are not all UTC; refunds are not always positive; bots are not filtered; transaction ids do not always join; FX rates skip weekends and 35 weekdays.

3. Key definitions and decisions

Production revenue. net_amount_ngn (gross $\times$ (1 – discount)) of every booking not Cancelled, by shoot date. Postponed and blank-status jobs are included, per the brief.

Foreign jobs. converted at the rate on the shoot date; weekends and missing days use the last published rate (38.9% of foreign-job rates were gap-filled). 7 $\times$ 10 typo rates corrected; the June 2023 and January 2024 devaluation jumps kept.

Merch revenue. quantity $\times$ price paid on lines whose order is not Cancelled, less refunds (capped at the order's value, allocated to lines by value). Refunds on cancelled orders are not deducted. Discount codes do not lower the price paid in the data, so discount value = list price – price paid.

Labour cost. day_rate_charged; 1,020 rows with no charged rate use the crew member's standard rate (the 0.5x/1x/1.5x multiplier is random with respect to overtime, so the standard rate is the unbiased estimate).

Genuine vs artefact loss. a job whose crew cost exceeds revenue is a genuine loss if it still loses money after removing crew-days where that person was also logged on another job and pricing estimated rates at 0.5x. 147 genuine, 35 artefacts.

Human session. user agent is not a crawler, script or SEO tool (23,054 bot sessions; every referral-spam session is also a bot).

Channel mapping. Google Ads = Google; Meta Ads = Facebook + Instagram; TikTok Ads = TikTok; YouTube Pre-roll = YouTube; Email / CRM = Email; Influencer has no utm source. Campaign 'always_on' = 'brand_always_on'.

Utilisation. distinct crew-days $\div$ (crew whose logged shifts span the week $\times$ 7).

Service type. 115 bookings have none. A model using price, crew and days predicted service type with only 42% accuracy, so they stay 'Unknown' rather than being guessed.

Complete logs: cleaning_report/CLEANING_DECISIONS.md and warehouse/WAREHOUSE_DECISIONS.md.

4. SQL answers (summary)

Q1 Quarterly revenue

quarter	production_revenue_ngn	merch_net_revenue_ngn
2023Q1	719,530,558	89,095,550
2023Q2	1,778,258,991	88,904,150
2023Q3	1,767,499,101	93,273,000
2023Q4	2,382,581,453	153,856,200
2024Q1	971,038,196	135,296,950
2024Q2	1,928,512,185	121,056,550
2024Q3	2,618,185,727	138,123,950
2024Q4	3,133,832,267	194,355,550

2025Q1	1,621,934,236	177,177,750
2025Q2	2,717,236,251	187,950,550

Q3 Worst-margin jobs

booking_code	service_name	status	net_revenue_ngn	labour_ngn	margin_ngn	margin_on_trusted_labour_ngn	verdict
BK-02437	Photoshoot	Completed	1,917,000	6,933,500	-5,016,500	-4,567,000	genuine_loss
BK-02103	Studio Rental	Completed	837,000	5,378,000	-4,541,000	-3,713,000	genuine_loss
BK-02309	Photoshoot	Completed	1,365,000	5,517,500	-4,152,500	-3,428,000	genuine_loss
BK-02632	Studio Rental	Completed	698,000	4,477,500	-3,779,500	-3,550,000	genuine_loss
BK-01616	Photoshoot	Completed	2,822,450	6,161,000	-3,338,550	-2,705,050	genuine_loss
BK-01966	Studio Rental	Invoiced	1,466,010	4,792,000	-3,325,990	-3,078,990	genuine_loss
BK-02145	Studio Rental	Completed	949,000	4,140,500	-3,191,500	-2,143,000	genuine_loss
BK-02988	Photoshoot	Completed	3,757,000	6,608,000	-2,851,000	-2,308,500	genuine_loss
BK-02682	Studio Rental	Invoiced	1,235,000	4,006,000	-2,771,000	-2,202,000	genuine_loss
BK-01304	Studio Rental	Completed	2,340,000	4,994,500	-2,654,500	-1,930,000	genuine_loss
BK-02198	Studio Rental	Unknown	2,108,000	4,359,000	-2,251,000	-625,500	genuine_loss
BK-01935	Studio Rental	Completed	1,240,000	3,358,000	-2,118,000	-1,112,500	genuine_loss
BK-02783	Unknown	Postponed	1,664,000	3,767,000	-2,103,000	-1,174,500	genuine_loss
BK-02427	Photoshoot	Completed	2,050,000	4,146,500	-2,096,500	-1,431,500	genuine_loss
BK-01291	Studio Rental	Postponed	959,500	3,029,000	-2,069,500	-1,414,000	genuine_loss
BK-02039	Studio Rental	In Production	249,000	2,291,500	-2,042,500	-1,346,000	genuine_loss
BK-02418	Photoshoot	In Production	3,433,000	5,386,500	-1,953,500	-774,000	genuine_loss
BK-01876	Studio Rental	Completed	523,800	2,378,500	-1,854,700	-1,590,700	genuine_loss
BK-01916	Studio Rental	Postponed	1,257,300	3,050,000	-1,792,700	-1,613,200	genuine_loss
BK-02369	Studio Rental	Completed	1,242,000	2,915,500	-1,673,500	-884,500	genuine_loss

Q4 Crew utilisation

Weeks above 85%: 0 of 130. Peak 20.2% (week of 2024-12-23), median 10.8% (11 edge weeks at the start and end of the data excluded from peak/median because the crew-span method undercounts active crew there). The binding role most often is Production Assistant. 824 crew-days were logged on two jobs at once; the worst week (2024-12-23) had 24.

Q5 FX exposure

year	invoice_currency	jobs	value_at_shoot_date_rate_ngn	value_at_2023_01_01_rate_ngn	devaluation_gain_ngn	pct_rates_gap_filled
2023	GBP	50	278,597,317	141,258,483	137,338,834	32.00

2023	USD	73	388,471,086	177,767,365	210,703,722	35.60
2023	all	123	667,068,403	319,025,848	348,042,556	34.10
2024	GBP	70	981,119,620	191,098,881	790,020,739	32.90
2024	USD	74	1,269,054,706	211,847,440	1,057,207,266	50.00
2024	all	144	2,250,174,325	402,946,321	1,847,228,005	41.70
2025	GBP	18	489,890,253	66,506,975	423,383,277	50.00
2025	USD	31	946,226,834	106,406,327	839,820,507	38.70
2025	all	49	1,436,117,087	172,913,302	1,263,203,785	42.90
all	all	316	4,353,359,816	894,885,470	3,458,474,345	38.90

Q6 Price-history integrity

check_name	result
raw: overlapping window pairs	29
raw: SKUs with at least one overlap	19
raw: windows with valid_to < valid_from	7
raw: SKU spellings -> SKUs	57 -> 45
core: overlapping window pairs	0
core: gaps between consecutive versions	0
core: SKUs without exactly one current version	0
core: EXCLUDE constraint present	1

Q7 Merch by collection

collection	units_sold	gross_ngn	discount_value_ngn	refunds_ngn	net_ngn	cogs_ngn	contribution_margin_ngn	contribution_margin_pct
Studio Basics	7,372	134,822,500	9,784,500	12,288,702	122,533,798	49,894,500	72,639,298	59.30
Lagos Nights	6,161	190,624,700	4,937,200	19,231,801	171,392,899	67,519,400	103,873,499	60.60
Anniversary '05	7,487	179,711,900	24,228,300	17,531,194	162,180,706	53,873,400	108,307,306	66.80
35mm Club	9,753	198,287,400	13,123,000	17,439,633	180,847,767	63,509,200	117,338,567	64.90
Behind The Lens	8,998	220,649,900	14,731,700	21,766,847	198,883,053	71,458,500	127,424,553	64.10
Golden Hour	8,286	282,530,400	32,061,400	26,545,217	255,985,183	105,199,300	150,785,883	58.90
Harmattan Drop	7,765	317,090,700	2,420,600	29,823,905	287,266,795	121,622,400	165,644,395	57.70

Q8 Refunds

Store refund rate 9.49% of sales value. SKUs above 2x the store average: 0. Highest: XT-OVE-042 Behind The Lens Oversized Tee at 12.43% (1.31x), dominant reason Changed mind.

reason	refunds	refund_amount_ngn	share_of_refunds_pct	on_returned_orders	on_cancelled_orders
Changed mind	1,793	60,728,250	20.80	883	910
Damaged	1,657	57,047,750	19.20	822	835
Wrong size	1,646	52,522,000	19.10	830	816
Late delivery	927	32,640,300	10.70	477	450
Not as pictured	894	29,985,450	10.40	415	479
Unknown	870	32,929,700	10.10	429	441
Duplicate order	844	30,119,550	9.80	420	424

Q9 Channel efficiency (all quarters)

channel_name	sessions	conversions	conversion_rate_pct	revenue_ngn	spend_ngn	cpc_ngn	cac_ngn	roas
Direct	18,168	389	2.14	36,108,400	0.00	–	–	–
Email	17,883	386	2.16	35,490,900	149,650,718	250	387,696	0.24
Google	24,077	533	2.21	50,383,100	154,874,562	241	290,571	0.33
Influencer	0	0	–	0.00	129,799,446	231	–	0.00
Meta	53,446	1,099	2.06	103,137,000	146,289,014	238	133,111	0.70
Referral / Other	36,126	759	2.10	71,985,000	0.00	–	–	–
TikTok	18,387	432	2.35	39,758,300	134,886,338	229	312,237	0.29
YouTube	11,905	253	2.12	24,307,800	144,535,639	245	571,287	0.17

Q10 Attribution gap

scope	converted_sessions	with_transaction_id	joined_to_real_order	pct_joined	session_revenue_ngn	unattributed_revenue_ngn	unattributed_share_of_session_revenue_pct	unattributed_as_pct_of_merch_net_revenue
all converted sessions	3,851	2,666	2,625	68.20	361,170,500	112,293,000	31.10	8.10
human converted sessions	3,851	2,666	2,625	68.20	361,170,500	112,293,000	31.10	8.10

Joined orders are a median 254 days from the session, so even the 'attributed' share is doubtful; treat ROAS as an upper-bound indicator of the web shop only.

Q11 Repeat purchase

43.6% of CRM buyers with a full 90-day window bought again within 90 days (827 of 1,896). By acquisition channel the rate ranges from 36.8% (Referral) to 47.1% (Unknown); chi-square = 8.75, p = 0.2713, so channel does not predict repeat buying. Guest checkouts (45.2% of lines) cannot be tracked.

Q12 Cross-sell

1,258 of 1,260 production clients also bought merch. Mean order: ₦57,574 for production clients vs ₦59,587 for merch-only; Welch t = -0.93, df = 8320.5, p = 0.352. No meaningful difference.

Q14 Monthly money flow

month	production_revenue_ngn	merch_net_revenue_ngn	labour_cost_ngn	ad_spend_ngn	merch_cogs_ngn	revenue_minus_known_cost_ngn	known_cost_pct_of_revenue
2023-01	165,179,583	27,690,050	24,713,000	26,366,846	13,274,800	128,514,987	33.40
2023-02	242,803,203	30,258,200	35,384,500	26,006,560	13,737,600	197,932,743	27.50
2023-03	311,547,772	31,147,300	51,695,500	24,437,431	14,297,100	252,265,041	26.40
2023-04	768,108,938	29,118,100	108,638,000	28,567,417	12,676,200	647,345,421	18.80
2023-05	408,972,817	26,475,150	62,271,000	25,252,925	11,978,100	335,945,942	22.90
2023-06	601,177,235	33,310,900	97,983,000	28,282,130	14,771,000	493,452,005	22.20
2023-07	610,151,431	31,809,000	94,921,500	26,342,105	13,559,000	507,137,826	21.00
2023-08	702,927,567	30,082,050	106,740,500	30,060,009	12,465,200	583,743,908	20.40
2023-09	454,420,104	31,381,950	68,957,500	33,473,604	13,779,900	369,591,050	23.90
2023-10	628,253,856	31,572,850	103,586,000	28,586,187	13,544,100	514,110,419	22.10
2023-11	963,219,714	65,494,050	121,351,500	29,468,848	26,699,100	851,194,316	17.30
2023-12	791,107,883	56,789,300	132,899,000	31,716,727	22,851,800	660,429,656	22.10
2024-01	162,712,806	44,568,450	22,546,000	32,030,061	18,339,300	134,365,895	35.20
2024-02	366,564,760	41,077,350	42,801,000	32,509,025	16,297,700	316,034,385	22.50
2024-03	441,760,630	49,651,150	69,734,000	28,924,110	20,271,800	372,481,870	24.20
2024-04	668,652,060	43,402,050	77,059,000	26,523,278	17,071,600	591,400,232	16.90
2024-05	520,071,834	37,973,400	84,760,500	30,946,378	15,024,500	427,313,856	23.40
2024-06	739,788,291	39,681,100	68,634,500	29,617,116	14,873,600	666,344,175	14.50
2024-07	880,176,616	48,154,300	100,147,500	30,419,298	17,323,000	780,441,117	15.90
2024-08	794,970,100	42,205,400	114,903,000	26,365,002	16,332,500	679,574,998	18.80
2024-09	943,039,012	47,764,250	78,344,000	26,076,932	17,505,700	868,876,630	12.30
2024-10	808,447,211	44,547,800	91,405,000	28,416,646	16,338,500	716,834,865	16.00
2024-11	1,128,825,985	72,778,450	123,555,500	29,699,554	26,215,700	1,022,133,681	14.90
2024-12	1,196,559,071	77,029,300	133,867,500	30,890,866	27,338,100	1,081,491,905	15.10

2025-01	271,737,823	54,859,100	25,333,500	28,079,743	19,017,600	254,166,080	22.20
2025-02	599,991,397	57,026,500	56,652,000	27,386,618	20,446,100	552,533,179	15.90
2025-03	750,205,017	65,292,150	65,947,000	34,459,384	22,489,000	692,601,783	15.10
2025-04	965,956,608	71,186,800	70,831,000	22,373,045	24,564,900	919,374,463	11.40
2025-05	1,119,756,014	56,813,450	72,724,000	26,896,359	19,464,500	1,057,484,605	10.10
2025-06	631,523,629	59,950,300	61,756,500	29,861,514	20,528,700	579,327,214	16.20

5. Limitations

- Costs outside crew day rates, ad spend and merch cost of goods are not in the data.
- The PostgreSQL instance used to build and verify the warehouse was local; warehouse/xtrim_warehouse.dump restores it (pg_restore) and dashboard/data/ lets the dashboard run without a database.
- Session revenue is what the analytics tool recorded; it covers 26.2% of merch net revenue (other sales go through Instagram, pop-ups and wholesale).
- Lead source cannot separate paid from organic social; the ad break-even figure in the memo is a threshold, not an estimate.